

1 Solution — GIAM 4.2

Problem 5

1.1 Problem

Determine two sets A and B such that both of the sentences $A \in B$ and $A \subseteq B$ are true.

1.2 Answer

$$A = \emptyset, \quad B = \{\emptyset\}$$

1.3 Verification

The claim is existential, so exhibiting one pair and checking both conditions is the whole proof.

$A \in B$. The set $B = \{\emptyset\}$ has exactly one element, written between its braces, and that element is $\emptyset$. Since $A = \emptyset$, we have $A \in B$.

*$A \subseteq B$. Every element of A must lie in B . But $A = \emptyset$ has no elements at all, so the requirement is satisfied **vacuously** — there is nothing that could violate it.*

Both were confirmed by direct evaluation. Note the two checks pass for completely unrelated reasons: the first because A is literally listed inside B , the second because A has nothing to contribute.

1.4 The Picture

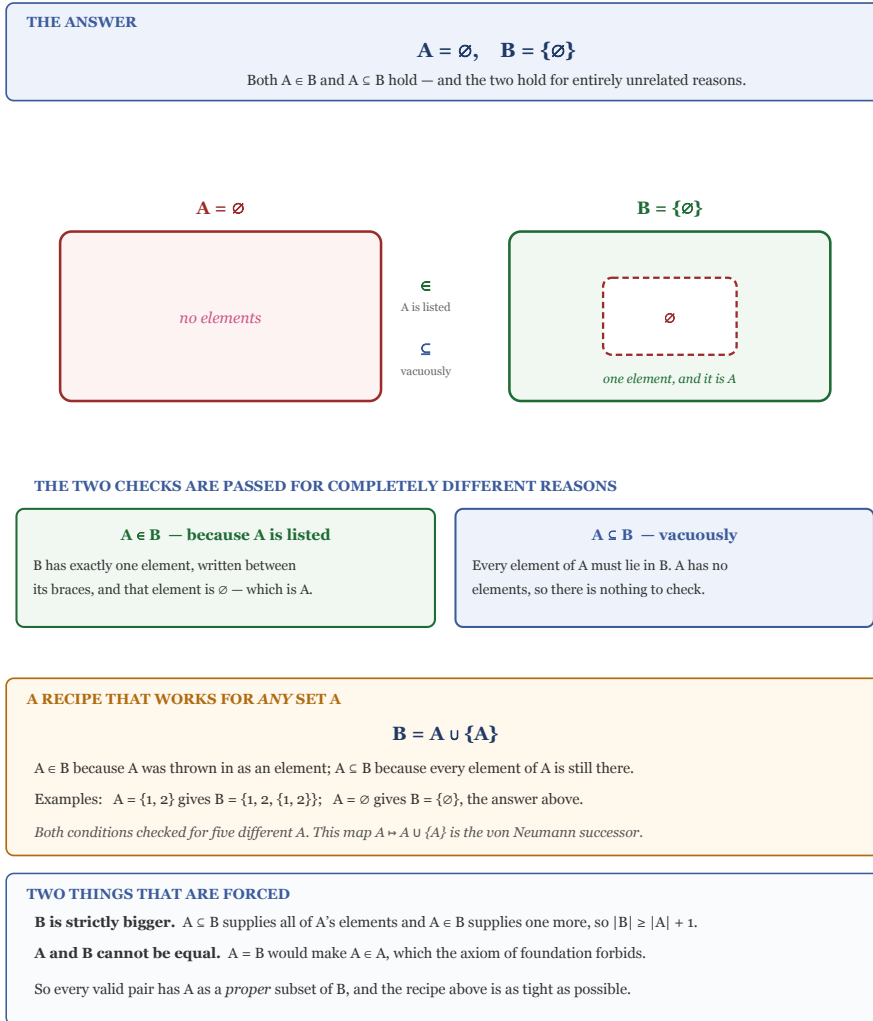


Figure 1.1: A diagram in four parts. The first gives the answer: A is the empty set and B is the set containing the empty set; both A is an element of B and A is a subset of B hold, and the two hold for entirely unrelated reasons. The second part draws two boxes. On the left, in red, a box labelled A equals the empty set, containing the words no elements. On the right, in green, a box labelled B equals the set containing the empty set, holding a smaller dashed box marked with the empty set symbol, with the caption that it has one element and that element is A. Between the boxes, an element symbol is drawn with the note A is listed, and below it a subset symbol with the note vacuously. Next, two cards explain that the checks pass for completely different reasons. The first, in green, headed A is an element of B because A is listed: B has exactly one element written between its braces, and that element is the empty set, which is A. The second, in blue, headed A is a subset of B vacuously: every element of A must lie in B, but A has no elements, so there is nothing to check. A band below gives a recipe that works for any set A: B equals A union the set containing A. A is an element of B because A was thrown in as an element, and A is a subset of B because every

element of A is still there. Examples: A equal to the set $1, 2$ gives B equal to the set $1, 2$, and the set $1, 2$; A equal to the empty set gives B equal to the set containing the empty set, the answer above. Both conditions were checked for five different A , and this map is the von Neumann successor. A final band gives two things that are forced. B is strictly bigger, since A being a subset supplies all of A 's elements and A being an element supplies one more, so the size of B is at least the size of A plus 1. And A and B cannot be equal, since that would make A an element of itself, which the axiom of foundation forbids. So every valid pair has A as a proper subset of B , and the recipe is as tight as possible.

1.5 Remark — A Recipe That Works for Any A

The answer above is the smallest example, but it is an instance of a construction that works universally. For **any** set A , put

$$B = A \cup \{A\}.$$

$A \in B$ because A was explicitly thrown in as an element.

$A \subseteq B$ because every element of A is still present — the union only added things.

Both conditions verified for five different choices of A . Some instances:

A	$B = A \cup \{A\}$
$\emptyset$	$\{\emptyset\}$
$\{\emptyset\}$	$\{\emptyset, \{\emptyset\}\}$
$\{1, 2\}$	$\{1, 2, \{1, 2\}\}$

Taking $A = \emptyset$ recovers the answer given. So the problem has not one solution but a solution for every set in mathematics.

The map $A \mapsto A \cup \{A\}$ is the **von Neumann successor**, the operation that builds the natural numbers from $\emptyset$:

$$0 = \emptyset, \quad 1 = \{0\}, \quad 2 = \{0, 1\}, \quad n + 1 = n \cup \{n\}.$$

So the answer to this exercise is precisely the statement $0 \in 1$ and $0 \subseteq 1$, and the second row of the table is $1 \in 2$ and $1 \subseteq 2$ — this problem is asking, without saying so, for a pair of consecutive natural numbers in the set-theoretic construction.

1.6 Remark — The Sets That Make This Easy Have a Name

A set B in which **every element is also a subset** is called **transitive**. For such a B , any element $A \in B$ automatically satisfies $A \subseteq B$ as well — so every element of a transitive set answers this problem.

That is exactly the property observed in 4.1 Problem 2, where each stage of the tower

$$\emptyset, \quad \{\emptyset\}, \quad \{\emptyset, \{\emptyset\}\}, \quad \dots$$

was found to be both an element and a subset of the next. Every von Neumann natural number is transitive, which is why the successor recipe never fails.

So rather than hunting for a lucky pair, one can say: **pick any transitive set B and any element A of it**. The exercise has a systematic answer rather than a clever one.

1.7 Remark — Two Things the Conditions Force

The pair cannot be arbitrary; the two requirements together pin down some structure.

B is strictly bigger than A . From $A \subseteq B$, every element of A lies in B . From $A \in B$, the object A is *also* in B , and it is not one of A 's own elements. So

$$|B| \geq |A| + 1,$$

confirmed on the examples above, where $|B|$ came out as exactly $|A| + 1$ each time. The successor recipe therefore produces the smallest possible B for a given A .

A and B cannot be equal. If $A = B$ then $A \in B$ becomes $A \in A$, which the **axiom of foundation** forbids — no set is a member of itself. So every valid pair has A as

a *proper* subset of B .

This is worth setting against 4.2 Problem 1, where each of the four blanks admitted exactly one of $\in$ and $\subseteq$. That was a feature of the particular sets chosen, not a law — the present problem exhibits pairs where both hold, and the previous remark shows there are as many such pairs as there are sets.

1.8 Remark — Why This Is Not a Contradiction

It is worth addressing the discomfort directly, because the two relations feel like they should be exclusive.

$\in$ and $\subseteq$ ask different questions, and nothing prevents both answers from being yes:

$$\begin{array}{l|l} A \in B & \text{is } A \text{ one of the objects listed in } B? \\ A \subseteq B & \text{is every object listed in } A \text{ also listed in } B? \end{array}$$

For $A = \emptyset$, $B = \{\emptyset\}$: the answer to the first is yes because $\emptyset$ appears in the list; the answer to the second is yes because the list for A is empty. The two questions never consult the same information, so their answers are logically independent.

What *is* true is that the relations behave differently: $\subseteq$ is transitive and reflexive, while $\in$ is neither. In particular $A \subseteq A$ always holds but $A \in A$ never does, which is exactly why $A = B$ was ruled out above.